

11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

- a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(\text{PRICE})$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?

a.

```
> library(POE5Rdata)
> library(AER) # ivreg() for 2SLS
> library(car) # linearHypothesis()
>
> data(fultonfish)
>
> # =====
> # Part (a):
> # =====
>
> reduced_form_a <- lm(lprice ~ mon + tue + wed + thu + stormy + mixed,
+ data = fultonfish)
> print(summary(reduced_form_a))
```

Call:

```
lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed,
    data = fultonfish)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.92281	-0.18819	0.00813	0.22404	0.68381

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.35957	0.07915	-4.543	1.50e-05 ***
mon	-0.10825	0.10339	-1.047	0.29753
tue	-0.06609	0.10103	-0.654	0.51446
wed	-0.04927	0.10377	-0.475	0.63591
thu	0.03935	0.10071	0.391	0.69681
stormy	0.44634	0.07921	5.635	1.51e-07 ***
mixed	0.23688	0.07851	3.017	0.00321 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3413 on 104 degrees of freedom

Multiple R-squared: 0.245, Adjusted R-squared: 0.2014

F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05

```
>
> f_test_stormy_mixed <- linearHypothesis(reduced_form_a,
+ c("stormy = 0", "mixed = 0"))
> print(f_test_stormy_mixed)
```

Linear hypothesis test:

stormy = 0

mixed = 0

Model 1: restricted model

Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Yes, MIXED is significant at 5% level.

The resulting F-value is greater than 10.

- b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.

```
> # =====
> # Part (b):
> # =====
>
> demand_2iv <- ivreg(lquan ~ lprice + mon + tue + wed + thu |
+                     mon + tue + wed + thu + stormy + mixed,
+                     data = fultonfish)
>
> print(summary(demand_2iv))
```

Call:

```
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
      tue + wed + thu + stormy + mixed, data = fultonfish)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-2.1327 -0.4320  0.0654  0.5133  1.1798
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533 < 2e-16 ***
lprice       -0.93014    0.35340  -2.632  0.00977 **
mon          -0.01190    0.20837  -0.057  0.95456
tue          -0.52583    0.20230  -2.599  0.01068 *
wed          -0.56262    0.20696  -2.719  0.00767 **
thu           0.09987    0.20285   0.492  0.62350
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom

Multiple R-Squared: 0.185, Adjusted R-squared: 0.1462

Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317

```
>
> vars  <- c("(Intercept)", "lprice", "mon", "tue", "wed", "thu")
> labels <- c("C", "ln(PRICE)", "MON", "TUE", "WED", "THU")
>
> tbl <- data.frame(
+   Variable      = labels,
+   Table_11.5_book = c(8.5059, -1.1194, -0.0254, -0.5308, -0.5664, 0.1093),
+   STORMY_MIXED_R = round(coef(demand_2iv)[vars], 4),
+   row.names = NULL
+ )
> print(tbl)
  Variable Table_11.5_book STORMY_MIXED_R
1      C      8.5059      8.5402
2 ln(PRICE)    -1.1194     -0.9301
3     MON     -0.0254     -0.0119
4     TUE     -0.5308     -0.5258
5     WED     -0.5664     -0.5626
6     THU      0.1093      0.0999
```

↑ The comparison.

- c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

C.

```
> # =====  
> # Part (c)  
> # =====  
>  
> resid_2sls <- residuals(demand_2iv)  
> sargan_aux <- lm(resid_2sls ~ mon + tue + wed + thu + stormy + mixed,  
+                   data = fultonfish)  
>  
> n <- nrow(fultonfish)  
> R2_sargan <- summary(sargan_aux)$r.squared  
> sargan_stat <- n * R2_sargan  
> df_sargan <- 1 # L - K = 2 instruments - 1 endogenous = 1  
> p_sargan <- pchisq(sargan_stat, df = df_sargan, lower.tail = FALSE)  
> print(p_sargan)  
[1] 0.3795467
```

Since the p-value is larger than 0.05, we accept that the variable MIXED is uncorrelated with the error.

- d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

```
d. > # =====  
> # Part (d)  
> # =====  
> f_test_days <- linearHypothesis(reduced_form_a,  
+                               c("mon = 0", "tue = 0",  
+                               "wed = 0", "thu = 0"))  
> print(f_test_days)
```

Linear hypothesis test:

```
mon = 0  
tue = 0  
wed = 0  
thu = 0
```

Model 1: restricted model

Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

```
>  
> f_days <- f_test_days$F[2]  
> print(f_days)  
[1] 0.6278532  
> p_days <- f_test_days$`Pr(>F)`[2]  
> print(p_days)  
[1] 0.6436925
```

The p-value of the test of joint significant is larger than 0.05, so we're unable to estimate the supply equation by 2SLS with confidence in our procedure.

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

a.

$$\text{Demand: } Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di} \quad (11.11)$$

$$\text{Supply: } Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si} \quad (11.12)$$

Rewrite 11.11 :

$$P_i = \underbrace{\left(\frac{-\alpha_1}{\alpha_2}\right)}_{a_1} + \underbrace{\left(\frac{1}{\alpha_2}\right)}_{a_2} Q_i + \underbrace{\left(\frac{-\alpha_3}{\alpha_2}\right)}_{a_3} PS_i + \underbrace{\left(\frac{-\alpha_4}{\alpha_2}\right)}_{a_4} DI_i + \left(\frac{-e_{di}}{\alpha_2}\right)$$

Rewrite 11.12 :

$$P_i = \underbrace{\left(\frac{-\beta_1}{\beta_2}\right)}_{b_1} + \underbrace{\left(\frac{1}{\beta_2}\right)}_{b_2} Q_i + \underbrace{\left(\frac{-\beta_3}{\beta_2}\right)}_{b_3} PF_i + \left(\frac{-e_{si}}{\beta_2}\right)$$

Parameters	a_2	a_3	a_4	b_2	b_3
Anticipated sign	-	+	+	+	+

- b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

b.

```
> # =====
> # Part (b)
> # =====
> inv_demand_2sls <- ivreg(p ~ q + ps + di | ps + di + pf,
+                           data = truffles)
>
> inv_supply_2sls <- ivreg(p ~ q + pf | pf + ps + di,
+                           data = truffles)
>
> print(summary(inv_demand_2sls))
```

Call:
ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)

Residuals:

	Min	1Q	Median	3Q	Max
	-39.661	-6.781	2.410	8.320	20.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.428	13.592	-0.841	0.40810
q	-2.671	1.175	-2.273	0.03154 *
ps	3.461	1.116	3.103	0.00458 **
di	13.390	2.747	4.875	4.68e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567, Adjusted R-squared: 0.5056
Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06

```
>
> print(summary(inv_supply_2sls))
```

Call:
ivreg(formula = p ~ q + pf | pf + ps + di, data = truffles)

Residuals:

	Min	1Q	Median	3Q	Max
	-9.7983	-2.3440	-0.6281	2.4350	11.1600

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-58.7982	5.8592	-10.04	1.32e-10 ***
q	2.9367	0.2158	13.61	1.32e-13 ***
pf	2.9585	0.1560	18.97	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486, Adjusted R-squared: 0.9448
Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16

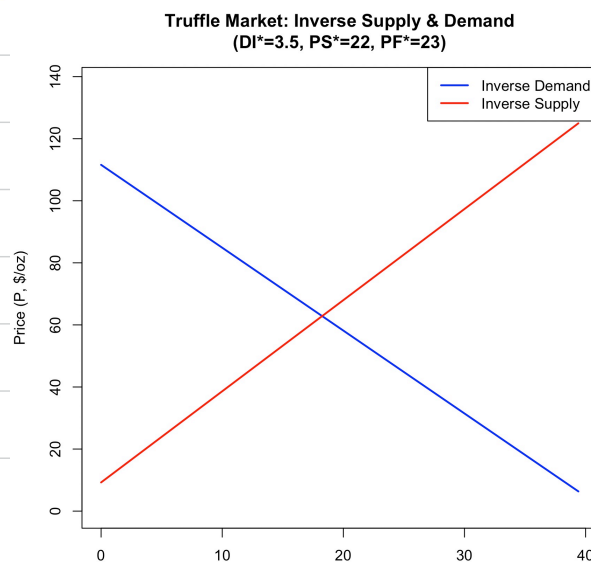
Yes, the signs are correct, and these estimated coefficients are significant ($p\text{-value} < 0.05$) different from zero.

- c. Estimate the price elasticity of demand “at the means” using the results from (b).

```
> # =====
> # Part (c)
> # =====
>
> d_coef <- coef(inv_demand_2sls)
> s_coef <- coef(inv_supply_2sls)
> a2_hat <- d_coef["q"]
> alpha2 <- 1 / a2_hat
> p_bar <- mean(truffles$p)
> q_bar <- mean(truffles$q)
> elasticity <- alpha2 * (p_bar / q_bar)
> print(elasticity)
      q
-1.272464
```

- d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.

```
> # =====
> # Part (d)
> # =====
>
> DI_star <- 3.5
> PF_star <- 23
> PS_star <- 22
>
> intercept_d <- d_coef["(Intercept)"] + d_coef["ps"] * PS_star + d_coef["di"] * DI_star
> slope_d <- d_coef["q"]
> intercept_s <- s_coef["(Intercept)"] + s_coef["pf"] * PF_star
> slope_s <- s_coef["q"]
>
> cat(sprintf("Inverse Demand (PS*=22, DI*=3.5) : P = %.4f + %.4f * Q\n",
+             intercept_d, slope_d))
Inverse Demand (PS*=22, DI*=3.5) : P = 111.5801 + -2.6705 * Q
> cat(sprintf("Inverse Supply (PF*=23) : P = %.4f + %.4f * Q\n\n",
+             intercept_s, slope_s))
Inverse Supply (PF*=23) : P = 9.2470 + 2.9367 * Q
>
> Q_range <- seq(0, max(truffles$q) * 1.5, length.out = 200)
> P_demand <- intercept_d + slope_d * Q_range
> P_supply <- intercept_s + slope_s * Q_range
>
> plot(Q_range, P_demand,
+      type = "l", col = "blue", lwd = 2,
+      xlim = c(0, max(Q_range)),
+      ylim = c(0, max(c(P_demand, P_supply), na.rm = TRUE) * 1.1),
+      xlab = "Quantity (Q, oz)", ylab = "Price (P, $/oz)",
+      main = "Truffle Market: Inverse Supply & Demand\n(DI*=3.5, PS*=22, PF*=23)")
> lines(Q_range, P_supply, col = "red", lwd = 2)
> legend("topright",
+      legend = c("Inverse Demand", "Inverse Supply"),
+      col = c("blue", "red"), lwd = 2)
```



- e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

e.

```
> # =====  
> # Part (e)  
> # =====  
>  
> Q_eq <- (intercept_s - intercept_d) / (slope_d - slope_s)  
> P_eq <- intercept_d + slope_d * Q_eq  
>  
> Q_rf_book <- 7.8951 + 0.6564 * PS_star + 2.1672 * DI_star - 0.5070 * PF_star  
> P_rf_book <- -32.5124 + 1.7081 * PS_star + 7.6025 * DI_star + 1.3539 * PF_star  
>  
> comparison <- data.frame(  
+   Method   = c("2SLS", "Table 11.2"),  
+   Q_hat    = round(c(Q_eq, Q_rf_book), 4),  
+   P_hat    = round(c(P_eq, P_rf_book), 4)  
+ )  
> print(comparison)
```

	Method	Q_hat	P_hat
(Intercept)	2SLS	18.2502	62.8426
	Table 11.2	18.2601	62.8142

The equilibrium values of P and Q obtained in (d) are 62.8426, 18.2502, respectively.

This agrees with the same value we obtain from Table 11.2, where the equilibrium values of P and Q are 62.8142 and 18.2601, respectively.

- f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

f.

```
> # =====
> # Part (f)
> # =====
> inv_demand_ols <- lm(p ~ q + ps + di, data = truffles)
> print(summary(inv_demand_ols))
```

Call:
lm(formula = p ~ q + ps + di, data = truffles)

Residuals:

	Min	1Q	Median	3Q	Max
	-25.0753	-2.7742	-0.4097	4.7079	17.4979

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-13.6195	9.0872	-1.499	0.1460
q	0.1512	0.4988	0.303	0.7642
ps	1.3607	0.5940	2.291	0.0303 *
di	12.3582	1.8254	6.770	3.48e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared: 0.8013, Adjusted R-squared: 0.7784
F-statistic: 34.95 on 3 and 26 DF, p-value: 2.842e-09

```
> inv_supply_ols <- lm(p ~ q + pf, data = truffles)
> print(summary(inv_supply_ols))
```

Call:
lm(formula = p ~ q + pf, data = truffles)

Residuals:

	Min	1Q	Median	3Q	Max
	-8.4721	-3.3287	0.1861	2.0785	10.7513

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-52.8763	5.0238	-10.53	4.68e-11 ***
q	2.6613	0.1712	15.54	5.42e-15 ***
pf	2.9217	0.1482	19.71	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared: 0.9531, Adjusted R-squared: 0.9496
F-statistic: 274.4 on 2 and 27 DF, p-value: < 2.2e-16

```
> d_ols <- coef(inv_demand_ols)
> d_2sl <- coef(inv_demand_2sls)
> demand_tbl <- data.frame(
+   Variable = c("Intercept", "Q", "PS", "DI"),
+   OLS       = round(d_ols, 4),
+   `2SLS`    = round(d_2sl, 4),
+   Diff      = round(d_2sl - d_ols, 4),
+   row.names = NULL
+ )
```

```
> print(demand_tbl)
  Variable      OLS      X2SLS      Diff
1 Intercept -13.6195 -11.4284  2.1911
2          Q   0.1512  -2.6705 -2.8217
3          PS   1.3607   3.4611  2.1004
4          DI  12.3582  13.3899  1.0317
```

```
> s_ols <- coef(inv_supply_ols)
> s_2sl <- coef(inv_supply_2sls)
> supply_tbl <- data.frame(
+   Variable = c("Intercept", "Q", "PF"),
+   OLS       = round(s_ols, 4),
+   `2SLS`    = round(s_2sl, 4),
+   Diff      = round(s_2sl - s_ols, 4),
+   row.names = NULL
+ )
```

```
> print(supply_tbl)
  Variable      OLS      X2SLS      Diff
1 Intercept -52.8763 -58.7982 -5.9219
2          Q   2.6613   2.9367  0.2754
3          PF   2.9217   2.9585  0.0368
```

Only the sign of q in the inverse demand equation, and it is also the only estimated coefficient that is not significant different from zero.

The difference between this result and part (b) is presented in the table to the right.